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ИНТЕГРАЛЬНЫЕ СПЕКТРЫ МГНОВЕННЫХ НЕЙТРОНОВ ДЕЛЕНИЯ ^{235}U И ^{239}Pu ТЕПЛОВЫМИ НЕЙТРОНАМИ

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Аннотация

Спектры мгновенных нейтронов деления ^{233}U , ^{239}Pu тепловыми нейтронами и спонтанного деления ^{252}Cf были измерены в идентичных экспериментальных условиях на исследовательском реакторе ПИЯФ (Гатчина). В процессе измерений идентичность геометрии эксперимента и последующего анализа данных, полученных для этих двух ядер, гарантировалась конструкцией экспериментальной установки и процедурой обработки данных. Измерения выполнялись одновременно для 11 значений угла между направлением эмиссии нейтрона и направлением вылета лёгкого осколка деления. Для селекции направления вылета осколка деления использовались многопроволочные пропорциональные счётчики. Энергия мгновенных нейтронов деления определялась с помощью метода времени пролёта. Для разделения событий от нейтронов и гамма-квантов дополнительно использовался метод формы импульса. В результате выполненного анализа данных были получены отношения интегральных спектров мгновенных нейтронов деления $^{233}\text{U}(n,f)$ и $^{239}\text{Pu}(n,f)$ к интегральному спектру мгновенных нейтронов спонтанного деления $^{252}\text{Cf}(sf)$. Полученные отношения имеют слабую зависимость от особенностей экспериментальной установки и свободны от погрешностей, присущих абсолютным измерениям. С использованием этих отношений и интегрального спектра мгновенных нейтронов деления $^{252}\text{Cf}(sf)$ в качестве рекомендованного международного стандарта, были вычислены интегральные спектры мгновенных нейтронов деления для $^{233}\text{U}(n,f)$ и $^{239}\text{Pu}(n,f)$.

Ключевые слова

деление ядер, ^{235}U , ^{239}Pu , тепловые нейтроны, спектры мгновенных нейтронов деления

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TOTAL PROMPT FISSION NEUTRON SPECTRUM FROM THERMAL NEUTRON-INDUCED FISSION OF ^{233}U AND ^{239}Pu

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Abstract

The prompt fission neutron spectra from thermal neutron – induced fission of ^{233}U , ^{239}Pu and spontaneous fission of ^{252}Cf have been measured under the equal experimental conditions at the PNPI research reactor (Gatchina). During the measurements, the identity of the experimental geometry and subsequent analysis of the data obtained for these two investigated nuclei was guaranteed by the construction of the experimental set-up and data processing procedure. The measurements were performed simultaneously for the 11 values of angle between neutron emission direction and the light fission fragment direction of motion. To select fission fragment direction, the multi-wire proportional counters were used. The prompt fission neutron energy was obtained by the time-of-flight method. The pulse shape analysis was additionally used for separation of neutron and γ -quanta events. As a result of the data analysis performed, the ratios of the total prompt fission neutron spectrum of $^{233}\text{U}(n, f)$ and $^{239}\text{Pu}(n, f)$ to the total prompt fission neutron spectrum of $^{252}\text{Cf}(sf)$ have been obtained. The obtained ratios have a weak dependence on the features of the experimental set-up and are free from errors inherent to absolute measurements. By using these ratios, the total prompt fission neutron spectra of $^{233}\text{U}(n, f)$ and $^{239}\text{Pu}(n, f)$ were calculated using the total prompt fission neutron spectrum of $^{252}\text{Cf}(sf)$ recommended as an international standard for neutron spectra measurement.

Keywords

nuclear fission, ^{235}U , ^{239}Pu , thermal neutrons, total prompt fission neutron spectra

Introduction

The total spectrum of prompt fission neutrons (PFNS) plays important role in many applications of the nuclear physics. For example, for prediction of the multiplication coefficient of the existing and newly developed power reactors it is widely used a variety of Monte-Carlo calculation methods where the PFNS is one of the main input parameters. At the recent Consultants' Meeting of the IAEA specialists [1], it was pointed to the necessity to carry out new evaluation of the PFNS both for main actinides $^{233,35,238}\text{U}$, ^{239}Pu and ^{232}Th and some minor actinides, such as ^{237}Np , ^{241}Am , $^{242\text{m}}\text{Am}$, ^{240}Pu and ^{245}Cm . In this connection, it was recommended to experimental groups to conduct new experiments or to present the results of formerly completed research works with full description of the applied corrections and analyzed uncertainties. The measurements of prompt fission neutron angular and energy distributions of $^{233}\text{U}(n, f)$, $^{239}\text{Pu}(n, f)$ and $^{252}\text{Cf}(sf)$ were performed at the equal experimental conditions by means of a set-up used earlier for analogous measurements [2-4]. It allowed to obtain the total prompt fission neutron spectra of $^{233}\text{U}(n, f)$ and $^{239}\text{Pu}(n, f)$ in the form of ratio of the total prompt fission neutron spectra of $^{233}\text{U}(n, f)$ and $^{239}\text{Pu}(n, f)$ to that of $^{252}\text{Cf}(sf)$, that enables to exclude the neutron detector efficiency uncertainty. This work presents the results of such investigations.

Experiment

The experiments for $^{233}\text{U}(n, f)$ were performed at the radial neutron channel №7 of the research reactor WWR-M (Gatchina) equipped with a low background reflector neutron guide 3 m in length. Neutron flux with average neutron energy of 0.0363 eV emerged from the neutron guide outlet slit (3 x 40 mm² in cross section) was equal to $\sim 2 \cdot 10^7 \text{ cm}^{-2} \cdot \text{sec}^{-1}$. The prompt neutron spectra measurements for $^{239}\text{Pu}(n, f)$ were carried out using the collimated neutron beam №1 of the WWR-M with a flux of $\sim 10^8$ thermal neutrons/cm² · sec. The ^{233}U and ^{239}Pu targets were fabricated by sputtering in vacuum on a 50 µg/cm² thick polyamide backing covered by 50 µg/cm² layer of gold, and on a 200 µg/cm² thick Al backing, respectively.

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The uranium target thickness was $280 \mu\text{g}/\text{cm}^2$ and made in the form of circular spot 15 mm in diameter. The plutonium target thickness was $150 \mu\text{g}/\text{cm}^2$ of the same shape and diameter. A ^{252}Cf layer 10 mm in diameter was made by self-transfer in vacuum on a 0.18 mm thick stainless steel foil. The fission fragment count rate was equal to $3 \times 10^4 \text{ sec}^{-1}$.

The experimental set-up used for these measurements is the same as in ref. [4] and is presented schematically in fig. 1. The prompt fission neutron spectra (PFNS) were measured simultaneously for eleven angles between the neutron and the fission fragment directions in the range from 0° to 180° in 18° intervals. To select fission fragment direction and to identify the fission event, the multi-wire proportional counters (MWPCs) were used. The start MWPC and target holder ring were placed on a special frame located in a centre of the chamber well out of the path of the neutron beam. The targets of the investigated nuclei were located in a centre of the chamber at the same place. The reaction chamber 350 mm in diameter was manufactured from 2 mm duraluminium and filled with isobutene up to the pressure of 4-5 Torr. The neutron beam was directed along the axis of the cylindrical reaction chamber normally to the disposal plane of all fission fragment and neutron detectors. The 16 rectangular stop MWPCs were installed in two arcing formations (8 MWPCs in each) diametrically opposed to each other in the reaction chamber at a distance of 140 mm from the chamber axis. The neutron beam was directed along the axis of the cylindrical reaction chamber normally to the disposal plane of all fission fragment and neutron detectors.

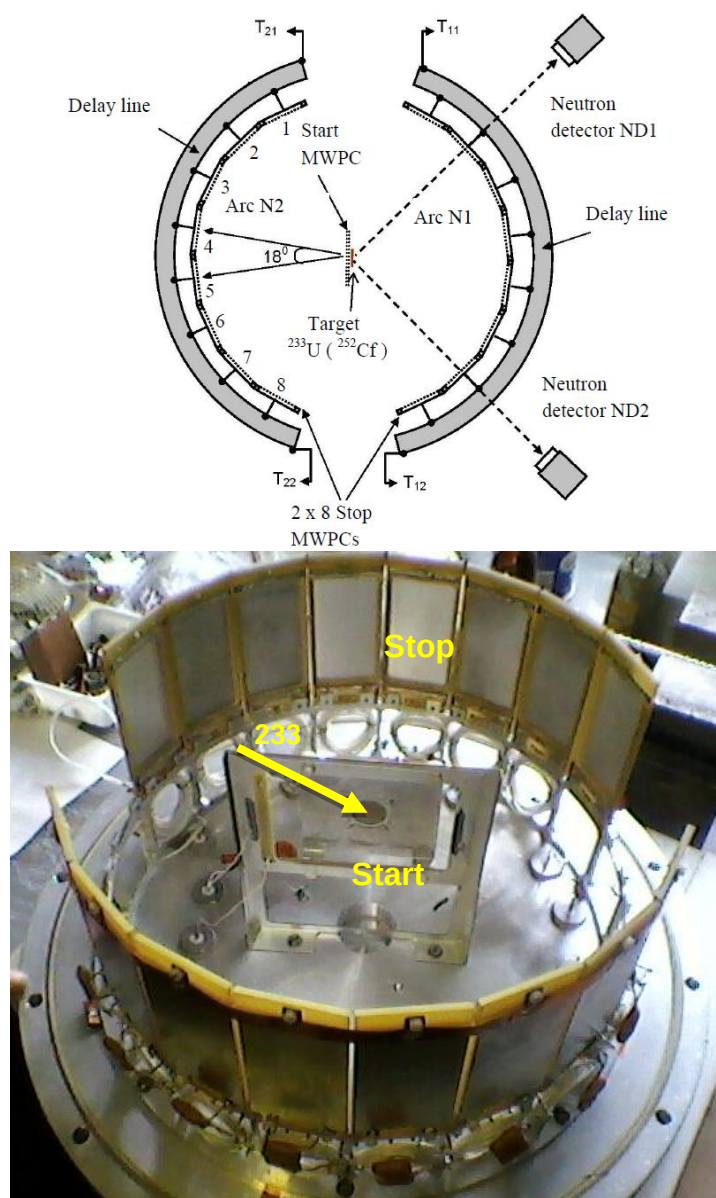


Fig. 1. Schematic view of the experimental set-up (no scale). The neutron-induced fission of ^{233}U and ^{239}Pu was directed normally to the drawing plane, along the reaction chamber axis. Right: reaction chamber

The prompt fission neutrons were detected by two neutron detectors (stilbene crystals $\varnothing 50 \times h 50$ mm and $\varnothing 60 \times h 40$ mm mounted on the Hamamatsu-R6091 photomultiplier). Neutron detectors were located at a distance of $d_1 = 49.2 \pm 0.3$ cm (solid angle 0.008 ± 0.001 steradian) и $d_2 = 47.2 \pm 0.3$ cm (solid angle 0.013 ± 0.001 steradian) from the target, respectively. The angle between the axes of neutron detectors was equal to 90° . Both neutron detectors were shielded by means of cylindrical shields made of a 30 mm thick layer of lead and 40 mm thick layer of polyethylene. Neutron registration threshold was 150-200 keV. A double-discrimination method (pulse shape and time-of-flight) was used to separate the events produced by neutrons and γ -quanta. The full time uncertainties of neutron detectors were defined from FWHM of the “fragment- γ -quantum” coincidence curve which was equal to $1 \div 1.2$ ns.

Three 2-weeks-long cycles of the measurements were carried out for every investigated nucleus. During every cycle, the angular and energy distributions of prompt fission neutrons were measured in turn for thermal neutron- induced fission $^{233}\text{U}(n,f)$, $^{239}\text{Pu}(n,f)$ and spontaneous fission of $^{252}\text{Cf}(s.f)$. Since $^{252}\text{Cf}(s.f)$ and $^{239}\text{Pu}(n, f)$ targets were prepared on the backing nontransparent for fission fragments, later on, for all investigated nuclei it were analyzed only the fission events registered by the stop MWPCs located on an Arc N2. For every measurement cycle the accumulated data were processed separately for each of two neutron detectors. The following corrections have been taken into account:

- time uncertainties in the time-of-flight measurements – the difference in average flight time of light and heavy fission fragments corresponding to the distance between target and start detector;
- effect due to correlation between the pulse-height of recorded signals and the position of the corresponding timing mark;
- neutron detectors background – the time independent accidental background, the neutron scattering inside the acceptance angle, residual γ -quanta background due to insufficient reliability of pulse shape separation of neutron and γ -quanta events (in the neutron energy range below 0.5 MeV) and components, connected with the availability of accidental coincidence between fragment and neutron belonging to different fission events;
- fission fragment detector efficiency;
- angular and neutron energy resolution of the experimental set-up;
- the bin-width correction due to the finite width of the energy step of a grid on which the prompt neutrons spectra were defined;
- normalization correction connected to the fact that in the measurements the experimental angular histograms were used instead of continuous distributions.

A description of the experimental method and the used data processing is omitted here since a full treatment is presented in ref. [2, 4].

Data processing

Nowadays, the total PFNS of $^{252}\text{Cf}(s.f)$, $N_{Std}(E_n)$, [5] is a commonly accepted international standard for prompt fission neutron spectra measurement. That is why, the measurements of the angular and energy distributions of the PFNS of $^{233}\text{U}(n,f)$, $^{239}\text{Pu}(n,f)$ and $^{252}\text{Cf}(s.f)$ were performed at the identical experimental conditions.

It should be mentioned that the ratios $R(E_n)$ of the total PFNS of $^{233}\text{U}(n,f)$ and $^{239}\text{Pu}(n,f)$ to the total PFNS of $^{252}\text{Cf}(s.f)$ can be obtained, which are not dependent on the uncertainties due to both the time-of-flight base uncertainty δd and the channel width of the used TDC. Since the spectra measurements were performed under the same experimental conditions and equipment, and the data processing was done equally for investigated nuclei, the effects of neutron scattering from the elements of experimental apparatus or/and the room have to be reduced significantly in the ratio obtained. Herewith, the ratio $R(E_n)$ is independent of systematic error of the neutron detector efficiency. It means that the ratio is free of the assumption about the shape of the total PFNS of $^{252}\text{Cf}(s.f)$, $N_{Std}(E_n)$. Therefore, the total PFNS of $^{233}\text{U}(n,f)$ and $^{239}\text{Pu}(n,f)$ were represented as the ratio $R(E_n)$ that allows to minimize the uncertainties of obtained data and to define desirable spectrum using a valid Cf-standard $N_{Std}(E_n)$:

$$R(E_n) = \frac{\Phi_x(E_n)}{\Phi_{Cf}(E_n)} = \frac{\Phi_x(E_n)}{N_{Std}(E_n)\nu_{Cf}} = F(E_n) \frac{n_x(E_n)}{n_{Cf}(E_n)}, \quad (1)$$

$$n_x(E_n) = 2\pi \int_0^\pi n_x(E_n, \theta) \sin(\theta) d\theta, \quad (2)$$

where $\Phi_x(E_n)$ ($x = \text{U}$ and Pu , Cf) is the true total PFNS; ν_{Cf} is a recommended value of the average number of prompt neutrons per fission event of $^{252}\text{Cf}(\text{s.f.})$ ($\nu_{\text{Cf}} = 3.759$; ENDF/B-VII); $n_x(E_n, \theta)$ ($x = \text{U}$ and Pu , Cf) is the PFNS of x nucleus corrected for time uncertainties, neutron detectors background and fission fragment detector efficiency; $F(E_n)$ is the total correction which should be applied to the ratio $R(E_n)$. The total correction $F(E_n)$ can be written as a product of multipliers, which are responsible for the effects mentioned above:

$$F(E_n) = f_{\text{res}}^E(E_n) f_{\text{res}}^\theta(E_n) I(E_n) B(E_n), \quad (3)$$

$$I(E_n) = \frac{I_x(E_n)}{I_{\text{Cf}}(E_n)} \quad B(E_n) = \frac{B_x(E_n)}{B_{\text{Cf}}(E_n)}, \quad (4)$$

where f_{res}^θ and $f_{\text{res}}^E(E_n)$ are coefficients which take into account the angular and energy resolution; $I(E_n)$ and $B(E_n)$ are the normalization and the bin-width corrections to the ratio, respectively. The normalization corrections to the investigated ratios were calculated similarly to ref. [4]. To estimate the bin-width correction and its uncertainty, the procedure analogous that of ref. [4] was used. Herewith, it was accepted that the total PFNS of $^{235}\text{U}(\text{n, f})$, $^{239}\text{Pu}(\text{n, f})$ and $^{252}\text{Cf}(\text{s.f.})$ can be represented by a Maxwell distribution with parameter T equal to 1.34 MeV, 1.382 MeV and 1.42 MeV, respectively [6].

The PFNS $n_x(E_n)$ ($x = \text{U}$ and Pu , Cf) were obtained independently for either of two neutron detectors as the weighted average of 3 independent measurement cycles:

$$n_{xk}(E_n) = \frac{\sum_{i=1}^m n_{xki}(E_n) \sigma_{xki}^{-2}}{\sum_{i=1}^m \sigma_{xki}^{-2}}, \quad \sigma_{xk}^{-2}(E_n) = \sum_{i=1}^m \sigma_{xki}^{-2}(E_n), \quad \sigma_{xki}^2(E_n) = \frac{\sum_{l=1}^{20} (n_{xki}^*(l, E_n) - n_{xki}(E_n))^2}{20}, \quad (5)$$

where i is a certain measurement cycle (amount of week cycles $m = 6$); k is the number of neutron detector; $\sigma_{xki}^2(E_n)$ is the error of the PFNS $n_{xki}(E_n)$ obtained within i -th measurement cycle for k -th neutron detector; $n_{xki}^*(l, E_n)$ is the l -th perturbed PFNS of the measured one $n_{xki}(l, E_n)$. Every l -th perturbed spectrum $n_{xki}^*(l, E_n)$ was obtained from the measured time-of-flight spectrum $n_{xki}(t_n, \theta)$ by the random variation of amount of the recorded events within achieved statistical accuracy $\pm (n_{xki}(l, t_n, \theta))^{1/2}$. It was assumed that the errors at different time-of-flight channels do not correlate and this variation was performed randomly. The errors determined by this way include both the statistical uncertainties and the uncertainties connected with the accuracy of determination of the “time zero” mark, and the operation instability of the electronic equipment.

To estimate the total measurement uncertainties which additionally includes the uncertainties associated with the normalization and resolution corrections, the procedure described above by Eq. (5) have been done with PFNS $n_{xki}(E_n)$ corrected for the normalization and resolution corrections. The corrections used for that were obtained independently for each measurement cycle analogous to ref. [4]. The total uncertainties obtained in this way are denoted as $\sigma_{x1}(E_n)$ and $\sigma_{x2}(E_n)$ for every of two neutron detectors, respectively. Further these uncertainties were used to calculate the uncertainty of the total PFNS. The final PFNS $n_U(E_n)$, $n_{\text{Pu}}(E_n)$ and $n_{\text{Cf}}(E_n)$ were obtained as the weighted average of data from two neutron detectors processed separately:

$$n_x(E_n) = \frac{n_{x1}(E_n) \sigma_{x1}^{-2}(E_n) + n_{x2}(E_n) \sigma_{x2}^{-2}(E_n)}{\sigma_{x1}^{-2}(E_n) + \sigma_{x2}^{-2}(E_n)}, \quad (6)$$

$$\sigma_x^{-2}(E_n) = \sigma_{x1}^{-2}(E_n) + \sigma_{x2}^{-2}(E_n), \quad D_x = \frac{\sigma_x(E_n)}{n_x(E_n)} \quad (7)$$

where D_x is a relative error of the PFNS measurement of $^{233}\text{U}(\text{n, f})$, $^{239}\text{Pu}(\text{n, f})$ ($x = \text{U}$, Pu) or $^{252}\text{Cf}(\text{s.f.})$ ($x = \text{Cf}$).

The total corrections of eq. (3), applied to measured ratio of the total PFNS of $^{233}\text{U}(\text{n, f})$ and $^{239}\text{Pu}(\text{n, f})$ to $^{252}\text{Cf}(\text{s.f.})$, are presented in Tables 1, 2. As it can be seen, the total correction $F(E_n)$ does not exceed 3% in 0.2 ÷ 10 MeV neutron energy range.

Results and discussion

The obtained total PFNS $^{233}\text{U}(\text{n, f})$, $\Phi_U(E_n)$, and $^{239}\text{Pu}(\text{n, f})$, $\Phi_{\text{Pu}}(E_n)$, were calculated using the Eq. (1). The numerical data are presented in Tables 1, 2. Since the neutron energy grid of the recommended total PFNS of $^{252}\text{Cf}(\text{s.f.})$ is different from the energy grid applied in this experiment, the linear approximation of recommended one defined in the energy grid applied in this experiment was used as a standard $N_{\text{Std}}(E_n)$. This

linear approximation is also presented in Tables 1, 2. The total relative errors of the obtained total PFNS D_ϕ were calculated assuming that all errors mentioned above are independent and the distribution of systematic errors are uniform:

$$D_\phi^2 = D_R^2 + D_{Std}^2 = D_{S+M}^2 + \frac{(D_{Bckg}^2 + D_{BW}^2 + D_{En}^2)}{3} + D_{Std}^2, \quad (8)$$

where $D_{S+M}^2 = D_x^2 + D_{Cf}^2$ ($x = U$ and Pu) is a relative errors including statistical and measurement uncertainties of the ratio of total PFNS measurements obtained using Eq. (7) for $^{233}U(n, f)$, $^{239}Pu(n, f)$ and $^{252}Cf(sf)$, respectively; D_{Std} is a relative accuracy of the $^{252}Cf(sf)$ standard $N_{Std}(E_n)$ [3]; D_{Bckg} is a relative maximum uncertainty of the ratio $R(E_n)$ caused by a systematic uncertainty in the background determination; D_{En} is a relative maximum uncertainty of the ratio $R(E_n)$ connected with the neutron energy grid uncertainty conditioned by the fact that the neutron detector length h is comparable with the time-of-flight base D ($h/D \sim 8\%$) [4].

The total PFNS of $^{233}U(n, f)$ and $^{239}Pu(n, f)$ are presented in Fig. 2,3 as ratio to the Maxwellian distribution with parameters $\nu_U = 2.49$, $\nu_{Pu} = 2.89$ (ENDF/B-VII) and $T_M = 1.34$ MeV and 1.38, respectively [6]:

$$\mu(E_n) = \frac{R(E_n) \cdot N_{Std}(E_n) \cdot \nu_{Cf}}{M(T_M, E_n) \cdot \nu_x} = \frac{\Phi_x(E_n)}{M(T_M, E_n) \cdot \nu_x}. \quad (8)$$

Table 1.

Total PFNS of $^{239}Pu(n, f)$

E_n , MeV	F	R	D_R %	D_{S+M} %	D_{Bckg} %	D_{BW} %	D_{En} %	$N_{Std} \times 10^4$ 1/MeV	D_{Std} %	$\Phi_{Pu} \times 10^4$ Neutr/MeV	D_ϕ %
10.860	0.9903	0.558	22.4	18.1	5.9	1.8	22.1	9.103	2.9	19.1	22.6
9.043	0.9768	0.712	15.9	12.1	2.7	1.5	17.7	30.7	2.2	82.2	16.1
7.647	0.9815	0.671	11.8	8.4	1.5	1.2	14.3	77.9	1.9	196.5	12.0
6.552	0.9791	0.710	9.3	6.3	1.0	1.0	11.8	160.8	1.6	428.9	9.5
5.677	0.9794	0.734	7.6	5.0	0.7	0.8	9.8	283.6	1.5	783.0	7.7
4.967	0.9793	0.769	6.2	4.0	0.4	0.7	8.2	444.5	1.4	1284.6	6.4
4.383	0.9804	0.767	5.2	3.3	0.3	0.6	6.9	640.0	1.3	1845.1	5.3
3.896	0.9799	0.755	4.4	2.8	0.2	0.5	5.8	859.1	1.2	2438.6	4.5
3.486	0.9817	0.785	3.9	2.6	0.2	0.4	4.9	1092	1.2	3221.0	4.0
3.137	0.9833	0.785	3.4	2.4	0.1	0.4	4.2	1331	1.2	3925.1	3.6
2.839	0.9842	0.768	3.1	2.3	0.1	0.3	3.5	1564	1.2	4513.2	3.3
2.581	0.9841	0.785	2.7	2.1	0.1	0.3	3.0	1785	1.2	5268.4	3.0
2.357	0.9860	0.812	2.6	2.1	0.1	0.2	2.5	1994	1.1	6086.5	2.8
2.161	0.9872	0.785	2.4	2.1	0.1	0.2	2.1	2187	1.1	6457.0	2.7
1.988	0.9864	0.798	2.3	2.1	0.1	0.2	1.7	2365	1.2	7091.3	2.6
1.835	0.9874	0.760	2.2	2.0	0.1	0.2	1.4	2524	1.2	7210.8	2.5
1.699	0.9877	0.759	2.2	2.1	0.1	0.1	1.1	2671	1.3	7622.9	2.6
1.578	0.9896	0.775	2.2	2.1	0.1	0.1	0.9	2792	1.2	8133.6	2.5
1.469	0.9900	0.788	2.1	2.1	0.1	0.1	0.7	2900	1.2	8587.3	2.5
1.372	0.9908	0.783	2.1	2.1	0.1	0.1	0.5	2995	1.2	8815.5	2.4
1.283	0.9920	0.785	2.2	2.2	0.1	0.1	0.3	3075	1.2	9069.4	2.5
1.203	0.9929	0.775	2.3	2.3	0.1	0.1	0.2	3147	1.2	9164.6	2.6
1.130	0.9929	0.774	2.3	2.3	0.1	0.1	0.1	3199	1.2	9305.6	2.6
1.064	0.9929	0.768	2.4	2.4	0.1	0.1	0.1	3243	1.3	9359.4	2.7
1.003	0.9936	0.742	2.5	2.5	0.1	0.1	0.1	3283	1.6	9151.5	3.0
0.948	0.9954	0.788	2.5	2.5	0.1	0.1	0.1	3311	1.7	9809.4	3.0
0.896	0.9956	0.778	2.6	2.6	0.1	0.1	0.1	3333	1.8	9741.8	3.2
0.849	0.9946	0.771	2.6	2.6	0.1	0.1	0.1	3352	1.8	9711.6	3.2
0.806	0.9947	0.777	2.7	2.7	0.1	0.1	0.1	3360	1.8	9814.4	3.2
0.766	0.9955	0.786	2.9	2.9	0.1	0.1	0.2	3360	1.8	9926.7	3.4
0.728	0.9955	0.770	3.1	3.1	0.1	0.1	0.3	3360	1.9	9719.0	3.6
0.694	0.9954	0.790	3.1	3.1	0.1	0.0	0.4	3360	1.8	9978.5	3.6
0.661	0.9944	0.761	3.2	3.2	0.1	0.0	0.6	3353	1.7	9588.5	3.6
0.631	0.9963	0.724	3.3	3.3	0.1	0.0	0.8	3341	1.6	9091.9	3.7

0.603	0.9960	0.743	3.5	3.5	0.1	0.0	1.0	3329	1.7	9302.7	3.9
0.577	0.9960	0.747	3.7	3.6	0.2	0.0	1.2	3318	1.8	9317.7	4.1
0.553	0.9949	0.734	4.0	3.9	0.2	0.0	1.5	3298	1.8	9103.1	4.4
0.530	0.9951	0.749	3.9	3.8	0.2	0.0	1.7	3280	1.8	9234.5	4.3
0.488	0.9958	0.782	3.1	2.8	0.2	0.1	2.3	3229	1.7	9496.9	3.5
0.434	0.9963	0.757	3.3	2.7	0.2	0.1	3.3	3164	1.7	8998.7	3.7
0.388	0.9965	0.731	4.8	4.1	0.2	0.1	4.4	3076	1.7	8452.6	5.1
0.349	0.9964	0.797	5.8	4.9	0.3	0.0	5.5	2998	1.7	8981.4	6.1
0.316	0.9975	0.741	6.4	5.1	0.3	0.0	6.7	2921	1.7	8133.7	6.6
0.287	0.9978	0.711	9.5	8.3	0.3	0.0	7.9	2840	1.8	7589.3	9.6
0.263	0.9978	0.708	10.9	9.6	0.3	0.0	9.0	2755	1.8	7328.3	11.1
0.241	0.9986	0.767	12.6	11.2	0.7	0.0	10.1	2675	1.9	7707.4	12.8

Table 2.

Total PFNS of $^{233}\text{U}(n, f)$

E_n , MeV	F	R	D_R %	D_{S+M} %	D_{Bckg} %	D_{BW} %	D_{En} %	N_{Std} 10^4 1/MeV	D_{Std} %	$\Phi_U \times 10^4$ Neutr/Me V	D_ϕ %
16.650	1.0075	0.3793	33.1	21.1	23.1	2.7	37.6	0.1812	15.6	0.25833	36.6
13.290	1.0020	0.4655	22.6	12.6	15.6	2.3	28.4	1.759	4.5	3.0781	23.0
10.860	0.9795	0.4339	16.0	9.0	5.9	1.9	22.1	9.103	2.9	14.846	16.3
9.043	0.9947	0.4691	12.6	7.2	2.7	1.5	17.7	30.7	2.2	54.140	12.8
7.647	0.9815	0.5313	9.7	5.1	1.5	1.3	14.3	77.9	1.9	155.58	9.9
6.552	0.9809	0.5614	7.9	3.9	1.0	1.0	11.8	160.8	1.6	339.32	8.1
5.677	0.9773	0.5988	6.7	3.6	0.7	0.8	9.8	283.6	1.5	638.37	6.9
4.967	0.9759	0.6074	5.4	2.6	0.4	0.7	8.2	444.5	1.4	1014.9	5.6
4.383	0.9755	0.6433	4.6	2.3	0.3	0.6	6.9	640.0	1.3	1547.5	4.8
3.896	0.9771	0.6501	3.9	2.0	0.2	0.5	5.8	859.1	1.2	2099.4	4.1
3.486	0.9796	0.6495	3.3	1.7	0.2	0.4	4.9	1092	1.2	2666.2	3.5
3.137	0.9782	0.6550	3.0	1.8	0.1	0.4	4.2	1331	1.2	3277.0	3.3
2.839	0.9795	0.6585	2.7	1.8	0.1	0.3	3.5	1564	1.2	3871.4	2.9
2.581	0.9811	0.6745	2.4	1.7	0.1	0.3	3.0	1785	1.2	4525.8	2.7
2.357	0.9819	0.6903	2.2	1.7	0.1	0.2	2.5	1994	1.1	5174.1	2.5
2.161	0.9825	0.6735	1.9	1.4	0.1	0.2	2.1	2187	1.1	5537.1	2.2
1.988	0.9847	0.6938	1.9	1.6	0.1	0.2	1.7	2365	1.2	6167.8	2.2
1.835	0.9858	0.6882	1.7	1.5	0.1	0.2	1.4	2524	1.2	6529.2	2.1
1.699	0.9871	0.6960	1.6	1.4	0.1	0.2	1.1	2671	1.3	6987.9	2.0
1.578	0.9884	0.6864	1.8	1.7	0.1	0.1	0.9	2792	1.2	7203.9	2.2
1.469	0.9885	0.6635	1.7	1.7	0.1	0.1	0.7	2900	1.2	7233.0	2.1
1.372	0.9895	0.6929	2.0	2.0	0.1	0.1	0.5	2995	1.2	7801.3	2.3
1.283	0.9902	0.6861	1.4	1.3	0.1	0.1	0.3	3075	1.2	7930.2	1.8
1.203	0.9901	0.6771	1.7	1.7	0.1	0.1	0.2	3147	1.2	8009.8	2.1
1.130	0.9912	0.6943	1.5	1.5	0.1	0.1	0.1	3199	1.2	8349.3	1.9
1.064	0.9913	0.6847	1.9	1.9	0.1	0.1	0.1	3243	1.3	8346.4	2.3
1.003	0.9935	0.6778	2.1	2.1	0.1	0.1	0.1	3283	1.6	8365.0	2.6
0.948	0.9957	0.6848	1.7	1.7	0.1	0.1	0.1	3311	1.7	8523.4	2.4
0.896	0.9960	0.6761	2.4	2.4	0.1	0.1	0.1	3333	1.8	8470.3	3.0
0.849	0.9964	0.6873	2.0	2.0	0.1	0.1	0.1	3352	1.8	8659.7	2.7
0.806	0.9961	0.6888	2.4	2.3	0.1	0.1	0.1	3360	1.8	8700.3	3.0
0.766	0.9958	0.6748	2.4	2.4	0.1	0.1	0.2	3360	1.8	8522.9	3.0
0.728	0.9959	0.7021	2.1	2.1	0.1	0.1	0.3	3360	1.9	8868.0	2.8
0.694	0.9957	0.6946	2.5	2.5	0.1	0.0	0.4	3360	1.8	8772.8	3.1
0.661	0.9957	0.6922	2.6	2.6	0.1	0.0	0.6	3353	1.7	8725.0	3.1
0.631	0.9958	0.6733	2.4	2.3	0.1	0.0	0.8	3341	1.6	8455.9	2.9

0.603	0.9945	0.6939	2.7	2.7	0.1	0.0	1.0	3329	1.7	8683.1	3.2
0.577	0.9946	0.6943	2.9	2.8	0.2	0.0	1.2	3318	1.8	8659.9	3.4
0.553	0.9956	0.6986	3.0	2.9	0.2	0.0	1.5	3298	1.8	8660.3	3.5
0.530	0.9956	0.7045	3.3	3.1	0.2	0.0	1.7	3280	1.8	8686.6	3.7
0.488	0.9965	0.7064	2.2	1.8	0.2	0.1	2.3	3229	1.7	8573.7	2.8
0.434	0.9965	0.6953	2.7	1.9	0.2	0.1	3.3	3164	1.7	8269.2	3.2
0.388	0.9962	0.6677	3.5	2.5	0.2	0.1	4.4	3076	1.7	7720.0	3.9
0.349	0.9966	0.7084	4.4	3.0	0.3	0.0	5.5	2998	1.7	7983.4	4.7
0.316	0.9969	0.7179	4.7	2.6	0.3	0.0	6.7	2921	1.7	7882.1	5.0
0.287	0.9974	0.6859	5.7	3.4	0.3	0.0	7.9	2840	1.8	7322.5	6.0
0.263	0.9975	0.7040	6.7	4.2	0.3	0.0	9.0	2755	1.8	7290.6	6.9
0.241	0.9978	0.6558	7.5	4.7	0.7	0.0	10.1	2675	1.9	6594.0	7.8
0.221	0.9972	0.7289	9.4	6.5	2.9	0.0	11.3	2598	2.1	7118.3	9.6

For comparison, the experimental data from Ref. [6, 7] and the ENDF/B-VII evaluation are also shown. All experimental spectra given in these figures were measured by the time-of-flight method and obtained from the total PFNS measured relative to the $^{252}\text{Cf(sf)}$.

As it can be seen in Fig. 2, 3, for these data and data of Ref.[6] the agreement in a shape of the total PFNS is observed within the experimental errors in the neutron energy range above 1 MeV. In the low neutron energy range from 0.1 to 1 MeV the uncertainties of the data increase especially for $^{233}\text{U}(n,f)$. Therefore, further experimental efforts are requested to define the total PFNS in the low neutron energy range.

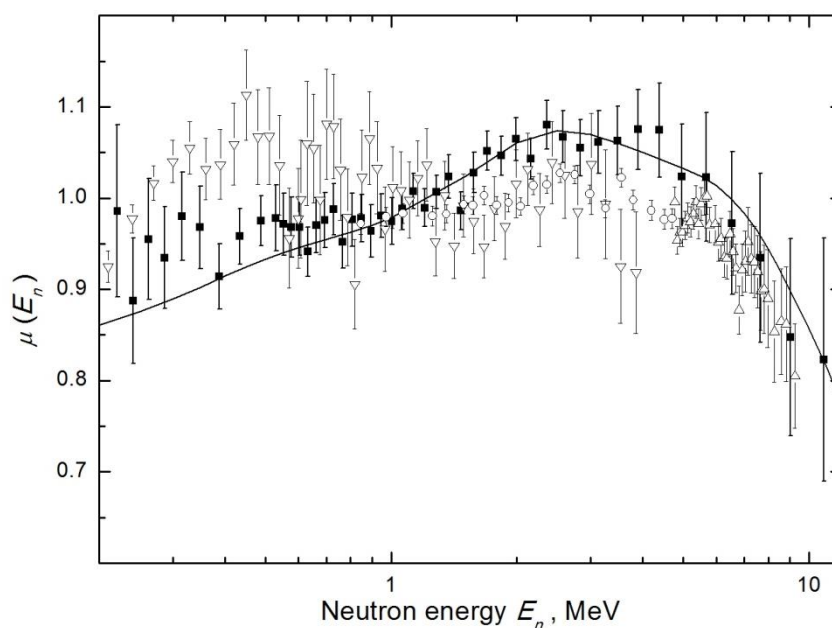


Fig. 2. Comparison of total PFNS of ^{233}U obtained by different experimental group and ENDF/B-VII. Data are presented as ratio to Maxwellian with $T=1.34$. All experimental data were normalized to $\nu_U=2.49$:

■ – present work; ○, Δ – [6]; ▽ – [7]; — – evaluation ENDF/B-VII

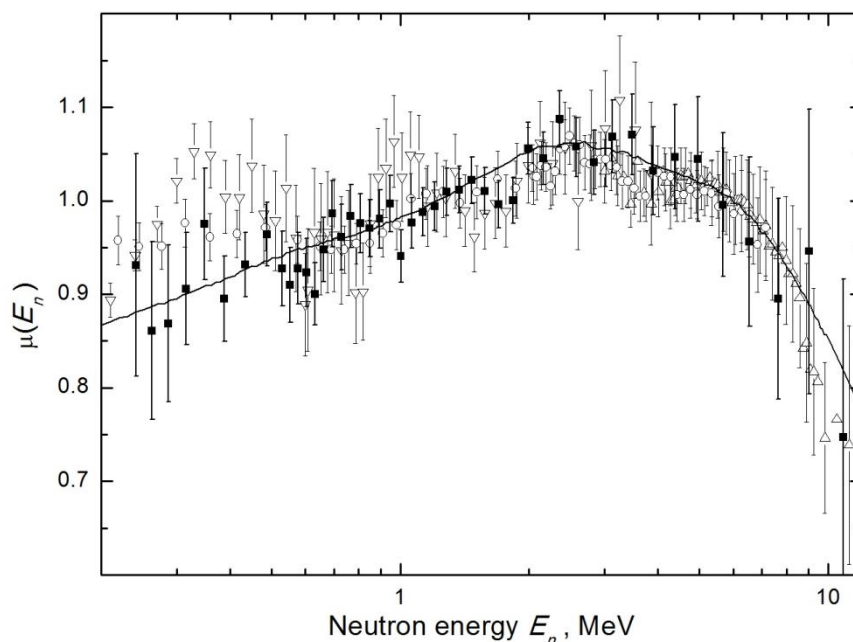


Fig. 3. Comparison of total PFNS of ^{239}Pu obtained by different experimental group and ENDF/B-VII. Data are presented as ratio to Maxwellian with $T=1.382$. All experimental data were normalized to $\nu_{\text{Pu}}=2.89$:

■ – present work; ○, Δ – [6]; ▽ – [7]; — – evaluation ENDF/B-VII

Conclusion.

The prompt fission neutron spectra from thermal neutron-induced fission of ^{233}U , ^{239}Pu and spontaneous fission of ^{252}Cf have been measured under the equal experimental conditions. The identity of the experimental geometry and analysis of the data obtained were guaranteed by the construction of the experimental set-up and data processing procedure. The measurements were performed simultaneously for the 11 values of angle between neutron emission direction and the light fission fragment direction of motion. The multi-wire proportional counters were used for registration of fission fragments while two stilbene scintillation counters were used to detect fission neutrons. The prompt fission neutron energy was obtained by the time-of-flight method. The pulse shape analysis was additionally used for separation of neutron and γ -quanta events. The ratios of the total prompt fission neutron spectrum of $^{233}\text{U}(n, f)$ and $^{239}\text{Pu}(n, f)$ to the total prompt fission neutron spectrum of $^{252}\text{Cf}(sf)$ have been obtained. The obtained ratios have a weak dependence on the features of the experimental set-up and are free from errors inherent to absolute measurements. By using these ratios, the total prompt fission neutron spectra of $^{235}\text{U}(n, f)$ and $^{239}\text{Pu}(n, f)$ were calculated using the total prompt fission neutron spectrum of $^{252}\text{Cf}(sf)$ as a standard.

The PFNS of ^{233}U and ^{239}Pu obtained in present measurements are in agreement with the evaluated data of ENDF/B-VII for neutron energies above 1 MeV. In the low neutron energy range from 0.1 to 1 MeV the uncertainties of the existing experimental data increase especially for $^{233}\text{U}(n, f)$. Therefore, further experimental efforts are needed to improve an accuracy of the total PFNS in the low neutron energy range.

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